

pos = 4 X

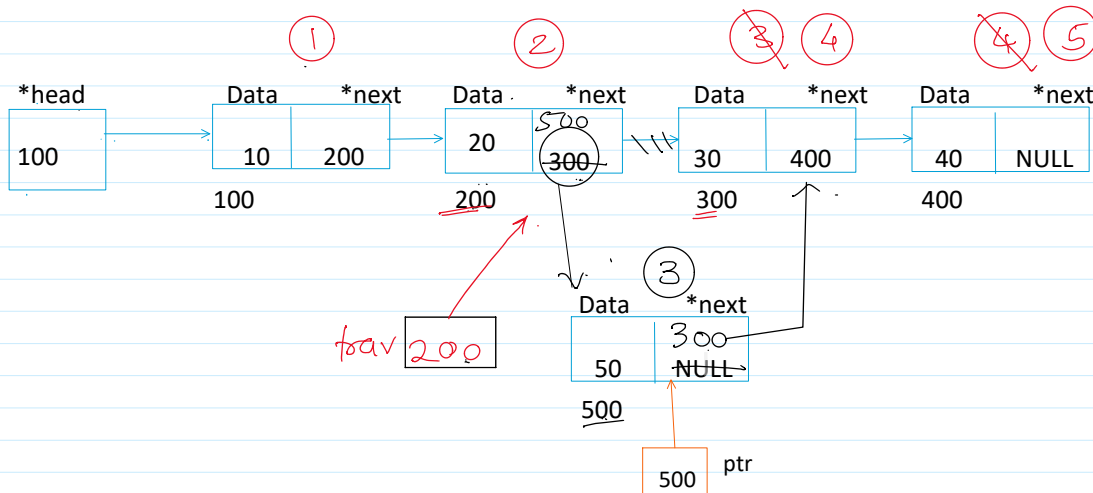
If List is empty

If (head == NULL)
 {
 If (pos == 1)
 addnode_at_first_pos(value);
 Else
 printf("Cannot add node at this pos");
 }
 If pos == 1
 addnode_at_1st_pos();
 If pos == count+1
 8 == 7+1
 addnode_at_last_pos();

If pos < 1 or pos > count+1

printf("Cannot add node at this pos");

- ① Create a node
- ② Assign the value
- ③ Attach.

pos → 3

1) Take a trav pointer starting from the first node.
Struct node *trav = head;

2) Traverse till pos-1 node.
For(int i=1; i<pos-1; i++)
 Trav = trav->next;

3) connect the new node to the current pos node(3rd node) of the linked list.
Ptr->next = trav->next;

4) Connect the pos-1 node to the new node.
Trav->next = ptr;

pos
 $500 \rightarrow \text{next} = 300;$
 $\text{ptr} \rightarrow \text{next} = \text{pos-1} \rightarrow \text{next};$
trav

$\text{ptr} \rightarrow \text{next} = \text{trav} \rightarrow \text{next};$
2nd →

$200 \rightarrow \text{next} = 500;$
 $\text{trav} \rightarrow \text{next} = \text{ptr};$